

Industrial Automation & Control

Week 3 Exam Notes

Lectures 9-14: Introduction to Automatic Control, P-I-D Control, and PID Tuning

Fast revision notes + concept explanations + likely exam answers

Focus Automatic control fundamentals, controller actions, PID, tuning	Exam angle Cheat-sheet style but explained from scratch	Based on Your topic list + Week 3 assignment reference questions on steady-state error, controller structure, phase lag, saturation, proportional band, derivative time, bumpless transfer, and velocity form
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1. Automatic Control - what it is and why we need it

Automatic control means making a system automatically adjust its input so that the output behaves the way we want. The controller compares what is desired (setpoint) with what is actually happening and then acts to reduce the difference called error.

- Main goals of automatic control: maintain stability, follow the setpoint, reject disturbances, reduce error, and improve speed of response.
- Setpoint or reference: desired value of output.
- Controlled variable: output we want to regulate.
- Manipulated variable: input changed by the controller.
- Disturbance: unwanted input that pushes the output away from the setpoint.
- Error: $e(t) = \text{setpoint} - \text{measured output}$.

Remember

- Good control does two things at the same time: it tracks the setpoint and fights disturbances.
- A controller that is fast but unstable is useless. A controller that is very stable but too slow is also poor.

2. Open-loop vs closed-loop control

This is one of the most important basics in the exam.

Feature	Open-loop control	Closed-loop control
Feedback	No feedback. Output is not compared with setpoint.	Uses feedback. Output is measured and compared with setpoint.
Disturbance handling	Cannot automatically correct	Can automatically reduce effect of

	disturbance.	disturbance.
Accuracy	Depends strongly on calibration and load conditions.	Usually better because error is corrected continuously.
Examples	Timer-based washing machine, toaster.	Speed control, temperature control, process control.

Key point: when the actuator saturates, the system effectively behaves like open-loop because the controller can no longer increase or decrease the actuator output further.

3. Stability and instability

- A stable system gives a bounded output for a bounded input and eventually settles.
- An unstable system shows growing oscillation or diverging output.
- Marginally stable systems oscillate without dying out.
- High loop gain improves error reduction, but too much gain together with phase lag can make the system unstable.

Phase lag is dangerous in feedback systems because the feedback signal becomes delayed in phase. If total phase lag approaches 180° while the loop gain is still high, negative feedback may act like positive feedback and oscillation can occur.

Likely exam point on phase lag

- Phase lag can come from time constants in the sensor, plant, and actuator, and from delays in transmitting/control action.
- More lag usually reduces stability margin.

4. Performance of a control system

Steady-state performance

- Concerned with final value after the transients die out.
- Main measure: steady-state error (SSE).
- Goal: make final error zero or very small.
- Integral action is the main way to eliminate constant-error bias.

Transient performance

- Concerned with how the response moves before settling.
- Main terms: rise time, overshoot, settling time, oscillation.
- Goal: small overshoot, fast rise, short settling time, no instability.
- Derivative action helps damping and improves transient behavior.

5. Important transient terms

- Rise time: time taken for output to rise from a small percentage to a large percentage of final value.
- Overshoot: amount by which output exceeds final value.
- Settling time: time taken to remain within a small band around final value.

- Steady-state error: final difference between setpoint and output.
- Bias input / load change / disturbance can create non-zero offset unless controller has integral action.

6. Why proportional control alone is not enough

Proportional control produces output proportional to error: $u(t) = K_p e(t)$. This is the simplest and most widely used control action.

- If error is large, controller output is large.
- If error becomes zero, proportional action becomes zero.
- Therefore proportional-only control cannot usually remove constant steady-state error completely in many practical systems.
- Increasing K_p reduces SSE and speeds up response, but too large K_p can increase overshoot and may destabilize the loop.

Effect of proportional gain K_p

- Higher K_p -> faster response and smaller SSE.
- Too much K_p -> larger overshoot, oscillation risk, less stability margin.
- Proportional band PB is inversely proportional to K_p .

7. P, I, and D actions explained from scratch

Mode	Law	Main benefit	Main problem	Best memory line
P	$u = K_p e$	Fast response, simple	Cannot remove SSE completely in many cases	Acts on present error
I	$u = K_i \int e \, dt$	Eliminates SSE for constant disturbances	Makes response slower, can increase oscillation	Acts on accumulated past error
D	$u = K_d \, de/dt$	Predicts trend, reduces overshoot, improves damping	Sensitive to noise; not used alone	Acts on rate of change of error

Complete ideal/parallel PID law: $u(t) = K_p e(t) + K_i \int e(t)dt + K_d \, de(t)/dt$

Industrial form often written as: $u(t) = K_p [e(t) + (1/T_i) \int e(t)dt + T_d \, de(t)/dt]$

From this form: integral coefficient = K_p/T_i . This is a common exam point and appears in your Week 3 assignment reference. Do not forget it.

8. Effects of each mode on transient and steady-state response

Controller mode	Steady-state error	Overshoot / oscillation	Speed notes
P	Reduces SSE but usually does not make it zero	May increase overshoot if K_p is high	Faster rise time
PI	Can make SSE zero for	Usually more overshoot	Slower than pure P but

	step/constant disturbance	than P alone if not tuned well	better accuracy
PD	Does not eliminate SSE by itself	Reduces overshoot and improves damping	Faster settling, better transient shape
PID	Can give zero SSE and good transient response if tuned well	Bad tuning can cause oscillation	Best compromise in many industrial loops

9. Proportional band, gain, integral time, derivative time

- Proportional gain K_p tells how strongly controller reacts to error.
- Proportional band (PB) is the error band required to move output through full range.
- PB is inversely proportional to K_p . Higher gain means smaller PB.
- Integral time T_i tells how fast the integral action accumulates.
- Smaller T_i means stronger integral action.
- Derivative time T_d tells the strength of derivative action.
- Derivative time is the time required for derivative action to equal proportional action for a ramp error. This is another direct exam point from the assignment.

Fast relation sheet

- K_p up -> PB down.
- T_i down -> integral action stronger.
- T_d up -> derivative action stronger, more damping, but more noise sensitivity.

10. Saturation and why it matters

Actuator saturation means the final control element has reached its maximum or minimum limit. Example: valve fully open, motor drive at maximum voltage.

- When saturation occurs, controller output demand may keep increasing but the actuator cannot respond further.
 - In that region the loop behaves like open-loop with respect to extra controller effort.
 - Integral term can continue accumulating error during saturation. This causes integral windup.
 - After saturation clears, the stored integral action can produce large overshoot and long settling.
- Anti-windup methods are often added in practice, although your main exam focus is likely just recognizing saturation and its effect.

11. Derivative control - practical implementation

Pure derivative on total error can cause a large kick when setpoint changes suddenly. In practice derivative action is normally applied to the incremental signal or measured variable trend, not directly to a sudden step in total error.

- Derivative is used mainly to improve damping and reduce overshoot.
- It is very sensitive to measurement noise because noise changes rapidly.

- Therefore derivative term is usually limited and filtered in practice.
- Common industrial practice: derivative on measurement / incremental signal to avoid output jumps during auto-manual transfer and setpoint steps.

Assignment-linked concept

- Your uploaded assignment states derivative block is normally applied to incremental signal Δu rather than total signal u . Reason: it helps prevent sudden jumps and supports bumpless transfer.
- Also, derivative term is often limited because high-frequency noise causes large rapid derivative output.

12. Bumpless auto-manual transfer

Auto-manual transfer means switching between automatic controller action and manual operator control. Bumpless transfer means this switching should not create a sudden jump or bump in controller output.

- In manual mode, operator directly sets the control output.
- When switching back to automatic mode, internal controller states must be initialized so that auto output matches current manual output at the switching instant.
- This is often achieved by adjusting the integral term or bias term before transfer.
- Goal: plant should continue smoothly without sudden output shock.

Bias input is useful here: controller output can be represented as bias + P + I + D contributions. During manual to auto transfer, the bias/integral state is aligned so that output remains continuous.

13. Digital implementation of PID controller

In digital control the controller runs at discrete sampling instants. Error is computed every sampling period and the controller output is updated numerically.

- Position form: controller directly computes absolute output $u(k)$.
- Velocity form or incremental form: controller computes change in output $\Delta u(k)$.
- The Week 3 assignment identifies incremental realization as velocity form.
- Digital derivative and integral are approximated using differences and sums.
- Sampling period must be chosen properly: too large makes control sluggish, too small amplifies noise/computation burden.

Typical discrete ideas: integral = sum of past errors, derivative = present error minus previous error divided by sampling time.

14. PID tuning - what tuning tries to achieve

PID tuning means choosing K_p , T_i , and T_d so the system is stable, accurate, and fast enough. Tuning is always a compromise between speed, damping, robustness, and actuator effort.

- Increase K_p to reduce error and speed up response, but not so much that the loop oscillates.
- Add integral action to eliminate SSE, but not so much that oscillation and long settling appear.
- Add derivative action to reduce overshoot and improve damping, but not so much that noise becomes a problem.

15. Practical tuning trends you should remember

- K_p too low $\rightarrow$ slow response, large SSE.
- K_p too high $\rightarrow$ oscillation, overshoot, instability risk.
- T_i too large (weak I) $\rightarrow$ SSE removed slowly.
- T_i too small (strong I) $\rightarrow$ hunting/oscillation and windup tendency.
- T_d too low $\rightarrow$ less damping benefit.
- T_d too high $\rightarrow$ noisy output, chattering actuator.

Tuning thumb rule

- Use P to get speed, I to kill offset, D to calm the response.
- This one line answers a lot of theory questions.

16. Likely descriptive exam answers

Question that can be asked	Compact answer to write
What is automatic control?	Automatic control is the use of feedback/controller action to make the output follow the desired setpoint, reject disturbances, and remain stable.
Differentiate open-loop and closed-loop control.	Open-loop has no feedback and cannot self-correct disturbances. Closed-loop uses feedback, compares output with setpoint, and reduces error automatically.
What is the significance of stability?	A control system is useful only if stable. Stability means the response remains bounded and settles after disturbance or input change.
What is steady-state error? How can it be reduced to zero?	SSE is the final difference between setpoint and output after transients die out. Integral action is used to remove constant SSE/offset.
Why is derivative control not used alone?	Derivative acts only on rate of change and gives no control for constant error; it is also noise-sensitive. So it is used with P or PI, not alone.
Why does derivative reduce overshoot?	Derivative anticipates the trend of error and adds damping, so the response slows before excessive overshoot develops.
Why is proportional band inversely related to K_p ?	A higher gain needs a smaller error to produce the same controller output swing, so proportional band decreases as gain increases.
What is bumpless transfer?	Bumpless transfer is switching between manual and automatic control without sudden jump in controller output, usually by matching internal bias/integral state to present output.

What is velocity form of PID?

Velocity form computes the change in controller output $\Delta u(k)$ instead of absolute output $u(k)$. It is the incremental realization used in digital controllers.

What happens during actuator saturation?

The actuator reaches its limit, extra control effort is ineffective, the loop behaves effectively like open-loop in that region, and integral windup may occur.

17. Assignment-linked concept notes (reference based)

Using the uploaded Week 3 assignment as reference, the likely exam focus areas are: steady-state error in closed loop, response with disturbance input, one/two/three degree-of-freedom controller structures, phase lag sources, tachogenerator effect on overshoot, actuator saturation, PID parameter meaning, proportional band vs gain, derivative time definition, bumpless transfer logic, derivative limitation due to noise, and incremental/velocity form realization. These are exactly the areas you should revise first.

18. Ultra-short final revision sheet

- Error = setpoint - output.
- Closed-loop beats open-loop for disturbance rejection.
- High loop gain reduces error but can hurt stability if phase lag is high.
- P reduces error and speeds response.
- I removes steady-state offset.
- D reduces overshoot and improves damping.
- PB is inversely proportional to K_p .
- Integral coefficient in industrial PID form = K_p/T_i .
- Derivative time: time for derivative action to equal proportional action for ramp error.
- Derivative is limited/filtered because noise creates large output spikes.
- Bumpless transfer = no jump when switching auto/manual.
- Incremental realization = velocity form.
- Actuator saturation $\rightarrow$ loop effectively behaves like open-loop there.

Prepared for last-minute exam revision. Keep this with your Week 3 assignment questions for practice.